Parallel Coupled Line Filter for 5G Communications

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Abstract- The purpose of this report is to design a microstrip parallel coupled line filter for 5G applications. There are three main steps in the design of the filter, the design of a Butterworth lowpass filter, the transformation to a lumped circuit bandpass filter and the transformation to microstrip parallel coupled line filter.

1. Introduction

In modern wireless communication technology, such as 5G communications, radio frequency (RF) noise has increasingly become a serious issue, thus the development and application of technology to filter the RF noise is essential (Hussaini et al., 2015). Filters have an important role in many RF microwave applications as they used to sperate or combine different frequencies (Hong & Lancaster, 2004). There is a high applicability for Microstrip band-pass filters in wireless communication technology as they are commonly used to filter unwanted frequencies and noise signals (Hong & Lancaster, 2004), particularly when applied to RF and microwave frequencies. To meet future requirements of 5G communication, it has been identified in which three bandwidths that 5G will operate at; low bandwidth spectrum at 700 MHz, 3.4-3.8 GHz with the potential to allow for wider bandwidths, and 26 GHz for ultra-dense very high-capacity networks (Ofcom, 2017). A bandpass filter contains a specific number of coupled resonators, the filter characteristics are defined by the dimensions of the distributed elements and the number of resonators, therefore, microstrip filter miniaturisation techniques involve the reduction of one of those defining quantities (Al-Yasir et al., 2018).

2. Literature Review

2.1. 5G communication

5G is the 5th generation mobile network. 5G has a higher performance and improved efficiency compared to the previous generations and is designed to enable anything and everyone to connect through a virtual network. 5G wireless technology is designed to deliver higher multi-Gbps peak data speeds, ultra-low latency, more reliability, massive network capacity, increased availability, and a more uniform user experience to more users (*What is 5g: Everything you need to know About 5G: 5G FAQ: Qualcomm*).

Wireless communication systems utilise radio frequencies to transmit information. 5G communications uses higher radio frequencies that are less cluttered, which are called ‘millimetre waves`, which means that more information can be carried and at a faster rate.

2.2. Commercial filters

Commercial filters are incredibly important in 5G communications. Commercial filters remove unwanted frequencies and noise, which are important when operating at these higher radio frequencies. Different filters have different parameters that will remove different frequencies depending of the design of the filter.

The bandpass filter (BPF) isolates a particular band frequency or filters out certain frequencies. A BPF only passes the frequencies within a certain desired band and attenuates others signals whose frequencies are either below a lower cut-off frequency or above an upper cut-off frequency. A typical bandpass filter can be obtained by combining a low-pass filter and a high-pass filter or applying conventional low pass to bandpass transformation

Due to its behaviour as a good resonator, a micro transmission line or strip line is being made as a filter. This microstrip line better compromises in size and performance rather other filters. In general, for manufacturing the microstrip circuit process, it is similar to the processes used in manufacturing printed circuit board. Furthermore, its advantage would be of largely being planar. The microstrip transmission lines consist of a conductive strip of width (W) and thickness (t) and ground plane, separated by a dielectric layer (ε) of thickness (h) as shown in Fig. 1 below

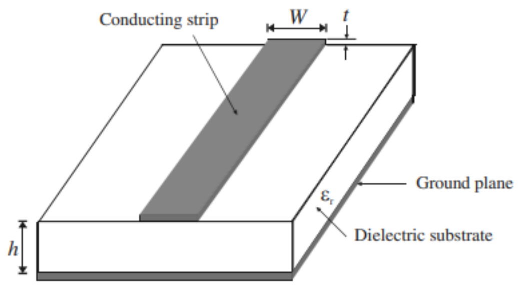

Fig. 1 – general microstrip structure

Referring to Fig. 3 below, a typical parallel coupled line bandpass filter which apply the half wavelength line resonators, whereas the resonators are parallel to each other along their length. This parallel coupled line structure with appropriate spacing between resonators caused a good coupling and provides desired bandwidth compared to another filter configuration. The coupled line section for θ is $\theta = \frac{\pi}{2}$, which will correspond to the center frequency of the bandpass response. Based on the Fig. 3, the width, gap, length, and impedance are labelled as (W), (S), (l), and (Y0) consequently.

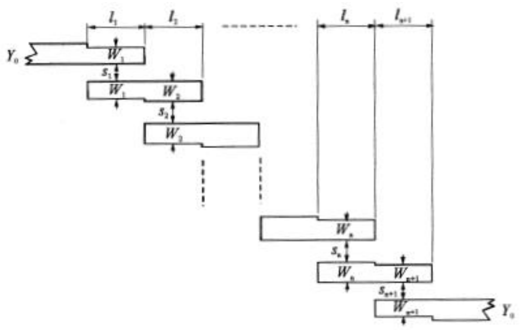

Fig. 2 – Common structure of microstrip parallel coupled line filter

The parallel coupling methods have certain advantages over the end coupling such as the length reduced approximately by half, a symmetrical insertion loss versus frequency response obtained with the first spurious response which occurred at three times centre frequency, the larger gap between strips allows for higher power rating filter and permits a broader bandwidth for a given tolerance

3. Filter Design and Configuration

The design procedure of the parallel coupled line filter includes the designing a low pass filter (LPF) prototype with normalised characteristics impedance (g_0) and cut-off frequency (Ω_c). From there, transformation techniques will be applied to convert the prototype LPF to a lumped component band pass filter to operate of the desired resonant frequency. Resulting in a BPF that comprises a circuit of lumped elements consisting of capacitors and inductors. By applying the Richards' transformation method to the band pass filter, it will convert the lumped components to a microstrip coupled parallel line filter.

3.1. Low Pass Filter prototype

A prototype 3-pole Butterworth low pass filter design in this section. The Filter will have a centre frequency of 3.6 GHz and a fractional bandwidth 11% and low ripple insertion loss. Figure 1 shows the equivalent circuit of the prototype low pass filter with low pass parameters g_i for $i = 0$ to $n + 1$.

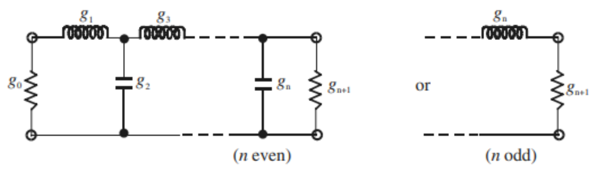

Fig. 3 – Low pass filter form with degree n

By applying Butterworth approximation to the low pass filter prototype, it will compute the element values of g_i, using these set of equations (1).

$$g_0 = 1, \quad g_i = 2\sin\{(2i-1)\frac{\pi}{2n}\} \quad g_{n+1} = 1 \tag{1}$$

As a result; $g_0 = g_4 = 1\mho$, $g_1 = g_3 = 1\mathrm{H}$ and $g_2 = 2F$ for $\Omega_c = 1.0\ rad/s$, from table 1.

Table 1 – Element values for Butterworth lowpass prototype filters

n	g_1	g_2	g_3	g_4	g_5	g_6	g_7	g_8	g_9	g_{10}
1	2.0000	1.0								
2	1.4142	1.4142	1.0							
3	1.0000	2.0000	1.0000	1.0						
4	0.7654	1.8478	1.8478	0.7654	1.0					
5	0.6180	1.6180	2.0000	1.6180	0.6180	1.0				
6	0.5176	1.4142	1.9318	1.9318	1.4142	0.5176	1.0			
7	0.4450	1.2470	1.8019	2.0000	1.8019	1.2470	0.4450	1.0		
8	0.3902	1.1111	1.6629	1.9616	1.9616	1.6629	1.1111	0.3902	1.0	
9	0.3473	1.0000	1.5321	1.8794	2.0000	1.8794	1.5321	1.0000	0.3473	1.0

3.2. Design of Lumped Component Band Pass Filter

To obtain the lumped element circuit of the band pass filter from the design low pass filter, which has a normalised source resistance/ conductance $g_0 = 1$ and we apply frequency and element $\Omega_c = 1.0\ rad/s$, we can apply frequency and element transformations techniques. The reactive elements are affected by the angular frequency conversion but has no effect on resistive components. The angular cut off frequency and the impedance scaling factor are given by (2).

$$\gamma_0 = 50 \qquad \omega_0 = \sqrt{\omega_2\omega_1} \tag{2}$$

Given the set of equations (3),

$$\begin{aligned} L_1' &= L_1 Z_0/\omega_0 FBW \\ C_1' &= FBW/\omega_0 C_1 Z_0 \\ L_2' &= FBW\,Z_0/\omega_0 L_2 \\ C_2' &= C_2/\omega_0 Z_0 FBW \\ L_3' &= L_3 Z_0/\omega_0 FBW \\ C_3' &= FBW/\omega_0 C_3 Z_0 \end{aligned} \tag{3}$$

Where $Z_0 = 50\Omega$ and $FBW = (\omega_2 - \omega_1)/\omega_0$, which the computed results are shown in table 2.

Table 2 – Element values lumped component bandpass filter prototype filters

Element	Value
L_1', L_3'	19.8915 nH
C_1', C_3'	0.0985 pF
L_2'	0.1232 nH
C_2'	15.9132 pF

Consequently, the equivalent circuit of lumped component band pass filter is shown below in Fig. 4.

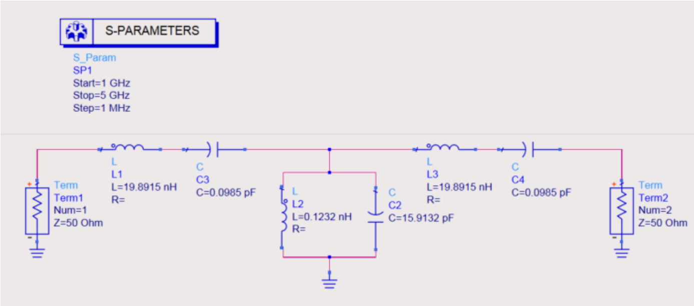

Fig. 4 – designed lumped component bandpass filter schematic

The data from the band pass filter is visualised in fig. 5. You can see that the bandwidth of the lumped component filter is about 3.4 GHz to about 3.8 GHz which satisfies the intended specifications.

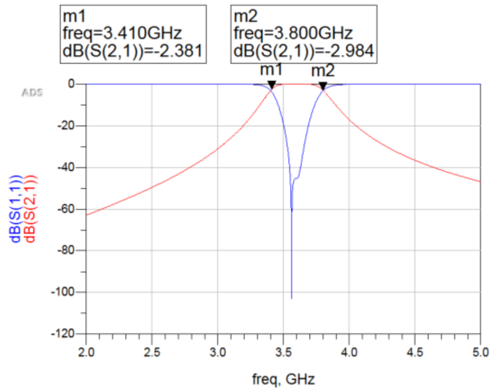

Fig. 5 – designed lumped component bandpass filter simulation

3.3. Design of the microstrip parallel coupled line filter

When designing the microstrip parallel coupled line filter, the values obtained previously through the other filters are used to compute the character impedance, which can be demonstrated through the set of equations (4).

$$Z_0J_1 = \sqrt{\frac{\pi FBW}{2g_0g_1}}$$
$$Z_0J_j = \frac{\pi FBW}{2\sqrt{g_kg_{k+1}}} \quad (4)$$
$$Z_0J_n = \sqrt{\frac{\pi FBW}{2g_ng_{n+1}}}$$

From the equations above, the set of equations (5) compute the even and odd type characteristic impedance of the parallel coupled microstrip line resonators below.

$$(Z_{oe})_j = Z_0[1 + J_jZ_0 + (J_jZ_0)^2] \quad (5)$$
$$(Z_{oo})_j = Z_0[1 - J_jZ_0 + (J_jZ_0)^2]$$

Where Z_{oe} is even characteristic impedance and Z_{oo} is odd characteristics impedance for j is from 0 to n. The calculated even and odd characteristics impedance is shown in table 3 below.

Table 3 – Characteristic Impedance (even and odd)

Stage	Z_{oe} (Ω)	Z_{oo} (Ω)
1	79.6478	37.8352
2	56.9453	44.5830
3	56.9453	44.5830
4	79.6478	37.8352

Using LineCalc in advanced design system (ADS), the dimensions of the width, spacing and length of each component are calculated using the computed values from above. These are shown in the below table 4.

Table 4 – Element width, length and spacing

Element	Width (mm)	Length (mm)	Spacing (mm)
CLIN1	6.196850	13.412165	2.772264
CLIN2	9.221220	12.967087	11.722402
CLIN3	9.221220	12.967087	11.722402
CLIN4	6.196850	13.412165	2.772264
50-Ohm Line	2.9	5	-

Fig. 6 shows the designed schematic of the microstrip parallel coupled line filter. A terminal at each end of the filter is placed so that the filter can be simulated, and the functionality of the filter can be observed.

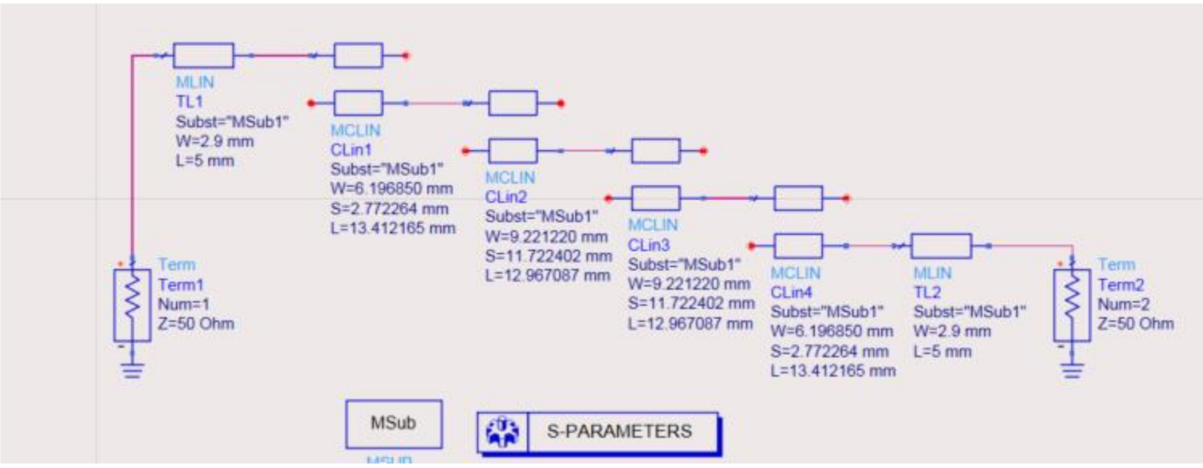

Fig. 6 – Designed microstrip parallel coupled line filter schematic

From ADS, the filter can be modelled which can be observed in fig. 7 below.

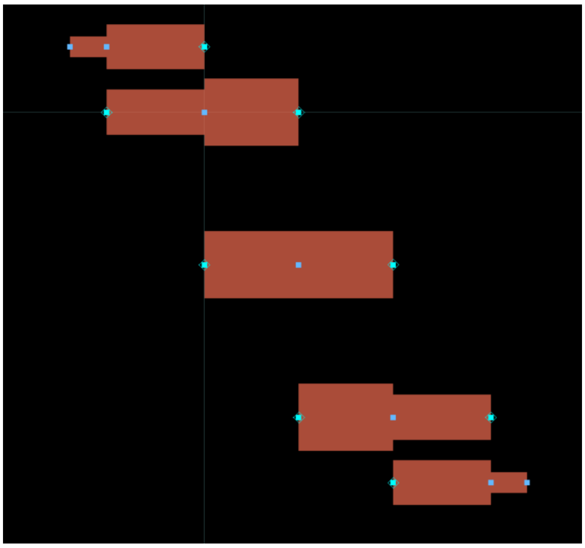

Fig. 7– Modelled microstrip parallel coupled line filter

4. Results analysis

Figure 8 below shows the results from the designed from the microstrip parallel coupled line filter.

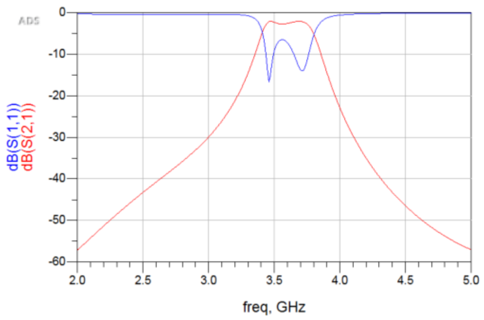

Fig. 8– Designed microstrip parallel coupled line filter simulation results

The results achieved by Advanced Design System software of the proposed microstrip parallel coupled line filter is depicted in fig. 8. The plot shows that the filter met the intended specifications, the bandwidth is approximately from 3.4 GHz to 3.8GHz and the centre frequency is about 3.6GHz which is as the design intended, for 5G communications. The results show the filter to be moderate in selectivity, a highly selective filter would have been more beneficial.

Through the ADS software, it is possible to simulate a 3D model which would represent the filter if it was to be fabricated, which can be seen in fig. 9.

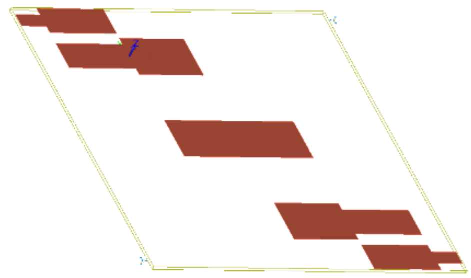

Fig. 9– 3D Modelled microstrip parallel coupled line filter

5. Conclusion

The S-Parameter characteristics of the simulation and measurement shows good filter performance. These simulations show that targeted specifications were met as a result. These specifications stated that the microstrip parallel coupled line filter would have a bandwidth of 3.4 to 3.8 GHz so that the filter could be used in 5G applications. From the simulation and results the proposed filter will theoretically function for 5G communications.

6. Reference List

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